SongSeeker: A Song Search Engine By Lyrics - Project Milestone

Jiachen Liang, Jiachen Liu, Joey Cai, Tong Tong
liang88@illinois.edu,jl315@illinois.edu,qcai20@illinois.edu,tong37@illinois.edu
University of Illinois Urbana-Champaign
Urbana, IL, USA

1 SUMMARY

SongSeeker aims to be a lyrics-based search engine that finds songs that match a user's input text, such as a mood, emotion, or thematic description Instead of depending only on literal keyword matching. In order to facilitate accurate and meaningful retrieval, we integrate both lexical and semantic representations.

We have finished dataset acquisition, analysis, and cleaning, and created a runnable local subset since the proposal. We have also implemented BM25 and Word2Vec retrieval systems. These elements serve as the foundation of a hybrid retrieval engine that can rank songs according to both semantic and lexical similarity.

Github repository: **SongSeeker**

2 INTRODUCTION

Traditional song-search systems have limited performance when the query is unclear or emotionally descriptive because they only return results that exactly match the words entered by the user. Our objective is to construct a lyric search engine, which allows users to freely describe scenes or emotions while still receiving song recommendations that are pertinent to their needs.

To achieve this, we integrate:

- Keyword-matching retrieval (BM25) – ensures high lexical precision.
- Semantic retrieval (Word2Vec, later BERT/SBERT) – captures conceptual similarity.

A hybrid ranking function will combine both signals to return balanced results.

3 DESCRIPTION

3.1 Data Preparation

We make use of the **Genius Song Lyrics Dataset** , which has approximately 5 million entries. The lyrics are cleaned up by deleting bracketed sections like [Chorus] and parenthesized asides (Remix), collapsing multiple spaces/tabs to one, trimming spaces around newlines, reducing three or more consecutive newlines to two, and stripping; non-string lyrics are left empty. Only the title, artist, tag, and lyrics are kept. Duplicates are eliminated based on the combination of title, artist, and lyrics, and entries with cleaned lyrics shorter than 20 characters are filtered out. For testing and debugging, a local subset of about 5,000 songs was extracted.

3.2 Exploratory Text Data Analysis

Table 1, Table 2, Table 3, Table 4 are some basic exploratory text data analysis of our lyrics dataset. In **Table 4**, all top frequency words are stop words and many words haven't been normalized. We need to further process the lyrics before we pass them into retrieval systems.

Feature	Count
Total lines (songs)	5134588
Unique artists	641318
Unique tags	6

Table 1: Basic Features Counts

Artist	Count
Genius Romanizations	16325
Genius English Translations	13831
Genius Brasil Tradues	8693
Genius Traducciones al Espaol	7083
Genius Traductions Franaises	4680
Genius Trke eviri	3941
Genius Russian Translations	3069
The Grateful Dead	2121
Genius Deutsche bersetzungen	1750
Genius Traduzioni Italiane	1657

Table 2: Top 10 Artists

Tag	Count
pop	2138467
rap	1724736
rock	793188
rb	196451
misc	181431
country	100315

Table 3: Top 10 Tags

Word	Count
the	38264898
I	34406255
a	23033729
you	22723668
to	21400070
me	15210353
and	14860830
my	13870986
in	13829294
of	12166312

Table 4: Top 10 Words (Stopwords are not dropped yet)

3.3 Lexical & Semantic Retrieval Models

We have implemented the BM25 keyword retrieval system and Word2Vec-based semantic text retrieval system. Both preprocesses each song's lyrics by converting to lowercase, removing punctuation from a defined set, and filtering out English stopwords.

Our BM25 model generates a document-term matrix that counts word occurrences per song and builds a vocabulary from all unique words. After applying the same preprocessing to a user query, the system calculates BM25 scores for each document using term frequency (TF), inverse document frequency (IDF), document length normalization (with tunable parameters k1=1.5 and b=0.75), and relevance scores.

Our semantic text retrieval system, Word2Vec, has two scoring modes: cosine similarity and average log-likelihood. It uses Gensim to load pre-trained word embeddings, caches mean vectors and per-document word vector matrices, and then calculates both similarity for every query.

Both models display the top five results along with the query's artists and song titles.

4 INITIAL EVALUATION

A complete evaluation of our model would require a labeled dataset to calculate all the metrics. Due to the large size of the dataset, even though we chose a subset of the entire dataset, it still takes a lot of time and we haven't finished yet.

Thus, for the initial evaluation we will choose the following five queries

- Q1: "love heartbreak"
- Q2: "party dance floor"
- Q3: "rain city night lonely morning fog forest quiet"
- Q4: "morning fog forest quiet"
- Q5: "space zero-gravity lonely exploration"

For each model and each query, we retrieve top 5 song from each model. Song relevance are manually checked to calculate precision@5 and MAP of each model. For Word2Vec, we used average log-likelihood to calculate similarity.

Table 5 and **Table 6** show the evaluation of two models, with BM25 has a higher MAP for these 5 queries. Again, this is only a preliminary evaluation, and the results may be influenced by subjective factors — for instance, how relevant the tester personally perceives a song to be. A more complete labeled test set and comprehensive test queries should be involved in the final evaluation.

Model	Q1	Q2	Q3	Q4	Q5
BM25	1.00	0.80	0.40	0.20	0.08
Word2Vec	0.60	0.60	0.80	0.20	0.20

Table 5: Precision@5 of 5 queries for both models

Model	Q1	Q2	Q3	Q4	Q5	MAP
BM25	1.00	1.00	0.75	0.20	1.00	0.65
Word2Vec	0.87	0.87	0.95	0.33	0.25	0.79

Table 6: Average Precision of 5 queries for both models

5 INSIGHTS SO FAR

Our experiments show that the dataset is clean and diverse enough for lyric-based retrieval. BM25 performs well for explicit keyword queries, while Word2Vec captures deeper semantic and emotional meanings. Manual checks on sample queries suggest the semantic model returns more intuitively relevant songs. These results support our plan to combine both approaches in a hybrid retrieval system for improved balance between precision and contextual relevance.

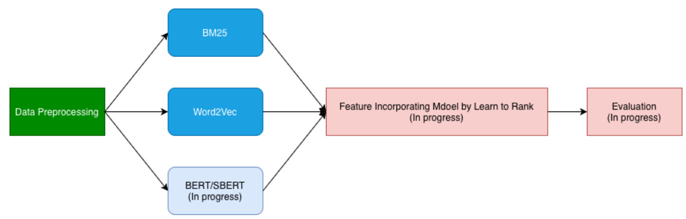

Figure 1: SongSeeker Project Workflow

Figure 1 shows the current project workflow we are following. Later on, we need to implement a semantic retrieval system based on BERT/SBERT embedding, a feature incorporation model by logistic regression and a complete evaluation. The latter two components require a labeled subset of the dataset, which will be created subsequently. If time permits, we would also explore optimizing techniques (e.g. parallel computing) to accelerate query execution, as current retrieval time of both models are very long (> 5 min).